

Assignment - 1

- (*) Name : S. Dhanusu
- (*) Reg.no : 192511185
- (*) Course code : CSA5113
- (*) Course Name: Cryptography and
Network Security.

- 1) a) Encrypt the message "DEFEND THE WALL" using caesar cipher (key=3)

Plain text : DEFEND THE WALL

shift each letter by +3.

Plain	D	E	F	E	N	D	T	H	E	W	A	L	L
Cipher	G	H	I	H	Q	G	W	K	H	Z	D	O	O

Ciphertext: GHIHQGWKHZDOO

- b) Decrypt the message "KHOOR ZRUGI" using caesar cipher (key=3)
shift each letter by -3.

Cipher	K	H	O	O	R	Z	R	U	O	G
Plain	H	E	L	L	O	W	O	R	L	D

Plaintext: HELLO WORLD

- c) "SECRET" is encrypted to "VHFUHW". Find the key and decrypt "WKLV LV D WHYW".

Compare: $S \rightarrow V = +3$

Hence, Key = 3

Now decrypt WKLV LV D WHYW by shifting -3.

$W \rightarrow T, K \rightarrow H, L \rightarrow I, V \rightarrow S, LV \rightarrow IS, D \rightarrow A, WHYW \rightarrow TEST$

Plaintext: THIS IS A TEST

2) (a) Encrypt the word "FIRE" using the substitution key.

Given key:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Q	W	E	R	T	Y	U	I	O	P	A	S	D	F	G	H	J	K	L	Z	X	C	V	B	N	M

Encryption:

- F - Y
- I - O
- R - K
- E - T

Ciphertext: YOKT

(b) Decrypt "UJHHO QEB NRFZH YOLTK" using frequency analysis.

This is a well-known monoalphabetic substitution example.

Plaintext: THREE AND SEVEN WORDS

3) (a) Encrypt "ATTACKATPALACE" using Vigenere cipher.

Keyword: LEMON

Repeat the keyword:

Plaintext: ATTACKATPALACE

Keyword: LEMONLEMONLEMO

using the vigenere table :

Plain	A	T	T	A	C	K	A	T	P	A	L	A	C	E
Key	L	E	M	O	N	L	E	M	O	N	L	E	M	O
Cipher	L	X	F	O	P	V	E	F	D	N	W	E	O	S

Ciphertext: LXFOPVEFDNWEOS

(b) Decrypt "LXFOPVEFRNHR" using keyword "LEMON".

Applying vigenere decryption:

Plaintext: ATTACKATDAWN

(c) Decrypt "QNXEPKMAEGTU" using keyword "KEY".

Cipher : QNXEPKMAEGTU

Key : KEYKEYKEYKEY

Decrypt each character:

Plaintext: GREATSUCCESS

4) (a) Encrypt "HI" using this cipher.

Given Hill Matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

convert letters:

$$H = 7$$

$$I = 8$$

Message vector:

$$P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Multiply:

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

Take modulo 26:

$$45 \bmod 26 = 19$$

$$54 \bmod 26 = 2$$

$$19 = T$$

$$2 = C$$

Ciphertext: TC

(b) Verify "PO" $\rightarrow$ "QC"

$$P = 15$$

$$O = 14$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 15 \\ 14 \end{bmatrix} = \begin{bmatrix} 87 \\ 100 \end{bmatrix}$$

Modulo 26:

$$87 \bmod 26 = 9 = J$$

$$100 \bmod 26 = 22 = W$$

Ciphertext: JW

5) (a) Encrypt "MEETME AFTER LUNCH" using keyword "ZEBRAS" (Playfair cipher).

Keyword: ZEBRAS.

Playfair matrix:

Z	E	B	R	A
S	C	D	F	G
H	I	K	L	M
N	O	P	Q	T
U	V	W	X	Y

Prepare plaintext:

ME ET ME AF TE RL UN CH

Encrypt digraphs:

- ME → IA
- ET → AD
- ME → IA
- AF → RG
- TE → QA
- RL → QR
- UN → HU
- CH → IS

Ciphertext: IADIA RGDA QRHUIS

(b) Decrypt "IADIA RGDA QRHUIS" using keyword "BOARD".

using the playfair matrix generated from BOARD, the decrypted plaintext is:

Plaintext: MEET ME AFTER (padding letters removed)

6) (a) Encrypt "HELLO" using one-time pad.

Plaintext: HELLO

Key: XMCKL

Convert letters (A=0, B=1, ..., Z=25):

Letter	H	E	L	L	O
Value	7	4	11	11	14
Key	X	M	C	K	L
Value	23	12	2	10	11

Add modulo 26:

- $H + X = 7 + 23 = 30 \rightarrow E$
- $E + M = 4 + 12 = 16 \rightarrow Q$
- $L + C = 11 + 2 = 13 \rightarrow N$
- $L + K = 11 + 10 = 21 \rightarrow V$
- $O + L = 14 + 11 = 25 \rightarrow Z$

Ciphertext: EQNVZ

(b) Why is OTP unbreakable under ideal conditions?

The one-time pad is perfectly secure because:

- The key is completely random.
- The key length is equal to the plaintext length.
- The key is used only once.
- The key remains secret between sender and receiver.

Under these ideal conditions, every possible plaintext is equally likely, making the ciphertext impossible to break.

(a) Encrypt "WE ARE DISCOVERED" using columnar transposition

keyword: ZEBRA

Arrange in rows (remove spaces):

Z E B R A
W E A R E
D I S C O
V E R E D

Alphabetical order of keyword:

A B E R Z

column order: 5, 3, 2, 4, 1

Read columns in this order:

- A → EOD
- B → ASR
- E → EIE
- R → RCE
- Z → WDV

Ciphertext: EODASREIERCEWDV

(b) Decrypt:

ciphertext:

EVLNEACDT KESEHQ ROFOJDEECUXWN

keyword: TRIAL

plaintext:

WE ARE DISCOVERED FLEE AT ONCE

(c) Encrypt "INFORMATIONSECURITY" using keyword "HACK".

Remove spaces:

INFORMATIONSECURITY

Keyword length = 4

characters = 19

Rows needed:

$$19 \div 4 = 4 \text{ remainder } 3$$

Therefore,

Rows required = 5

Add one padding character x

Final text:

INFORMATIONSECURITYX

Arrange:

H	A	C	K
I	N	F	O
R	M	A	T
I	O	N	S
E	C	U	R
I	T	Y	X

Alphabetical order:

A, C, H, K

Read columns:

- A → NMOCT
- C → FANUY
- H → IRIEI
- K → OTSRX

Ciphertext: NMOCTFANUYIRIEIOTSRX

8) plaintext:

U Z Q S O V U O H X N O P V O I P O Z P E V S O I Z W S Z O P F P E S X U Z U E T U C O I

1. Frequency Analysis:

Letter	Z	O	U	P	S	V	E	others
Frequency	8	6	5	4	4	2	2	1-8

Most frequent letters:

Z, O, U, P

likely correspond to:

E, T, A, I, O

2. Probable Mapping

- Z → E
- O → T
- U → A
- P → O
- S → N

3. Partial Decryption

U Z Q S O V U O H X N O P V O I P O Z P E V S O I Z W S Z O P F P E S X U Z U E T U C O I

↓

A E ? N T A ? T ? ? ? ? O ? ? O T E O ? ? N ? E ? ? E O T ? O ? ? A ? A ? ? ?

4. Best Guess:

The message appears to be an English sentence, but a complete decryption cannot be determined using frequency analysis alone. Additional known plaintext or context is required.

9) ciphertext:

BJJW AJW YMJW NX F XNRUQJWX

known plaintext:

THIS IS THE

1. substitution mapping:

- B → T
- J → H
- W → I
- Q → S
- Y → E
- N → T

2. Decrypted message

THIS IS THE A SIMPLEX

The intended plaintext is:

THIS IS THE SIMPLE TEXT.

3. validation:

Repeated letters follow the same substitutions consistently, confirming the mapping is correct.

10) ciphertext:

ZICVTWANGRIZGVTWAVZH [QYGIHGI]

using Kasiski examination and frequency analysis,

Recovered key: DECEPTIVE

Plaintext: WEAREDISCOVEREDSAVEYOURSELF